

This rule looks for the *USER* command in FTP traffic, indicating a login attempt. When detected, it sets a flowbit (*ftp_user_seen*) to track the occurrence of this action.

- *Detecting a file upload after login*

```
alert tcp any any -> any 21 (msg:"FTP
STOR after USER login"; content:"STOR
"; nocase; flowbits:isset,ftp_user_
seen; flowbits:unset,ftp_user_seen;
sid:500002;)
```

This rule triggers only if the flowbit *ftp_user_seen* is already set, meaning a *USER* command has been observed (a login attempt). It then looks for the *STOR* command, which indicates a file upload. Once fired, the flowbit is unset to reset the state for the next sequence.

The static rule for FTP traffic detection is always active and triggers whenever communication occurs on port 21, continuously monitoring from session start to exit. In contrast, the dynamic rules are context-aware and depend on specific actions in sequence: one detects a login attempt with the *USER* command and sets a flowbit, while the next triggers only if a file upload (*STOR* command) follows that login. This demonstrates multi-step detection, where alerts fire based on the logical flow of FTP activity rather than every packet.

Next, save the file using *Ctrl+O* and exit with *Ctrl+X*. Now, validate the *snort.conf* file using `$ sudo snort -T -i enp0s3 -c /etc/snort/snort.conf`.

Testing traffic

The traffic is tested using the console mode of the Snort. Listen to the *ftp* command in quiet mode by using this command:

```
$ sudo snort -A console -q -i enp0s3 -c
/etc/snort/snort.conf
```

To test the FTP traffic, an FTP connection request must be sent from another system on the same network (for example, a Windows machine) to

Figure 4: Writing custom rules to detect FTP traffic

Figure 5: Successful execution of *ftp* command triggering the static rule from a Windows system

Figure 6: Successful listing of alert messages in the Snort console

Figure 7: Successful execution of the *ftp* command from a Windows system